

The alkene region and the benzene region

In ^{13}C NMR, alkene and benzene carbons came in the same region of the spectrum, but in the ^1H NMR spectrum the H atoms attached to arene C and alkene C atoms sort themselves into two groups. Look at the ^{13}C and ^1H chemical shifts of cyclohexene and benzene. The two carbon signals are almost the same (1.3 ppm difference, < 1% of the total 200 ppm scale) but the proton signals are very different (1.6 ppm difference = 16% of the 10 ppm scale).

Molecule	^{13}C Shift (ppm)	^1H Shift (ppm)	$\Delta(^{13}\text{C})/\text{Scale}$	$\Delta(^1\text{H})/\text{Scale}$
Benzene	128.5	7.3	-	-
Ethylene	123.3	5.3	2.5%	20%

1. An organic compound ($C_7H_{12}O_2$) exhibited the following data in the 1H NMR spectrum.

δ 7.10 (1H, dt, $J = 16$ Hz and 7.2 Hz), 5.90 (1H, dt, $J = 16$ Hz and 2 Hz), 4.1 (2H, q, $J = 7.2$ Hz), 2.10 (2H, m), 1.25 (3H, t, $J = 7.2$ Hz), 0.90 (3H, t, $J = 7.2$ Hz) ppm. (CSIR NET)

Q2. Determine the correct structure of a compound having molecular formula $C_6H_{11}BrO_2$, based on the following characteristic spectral data:

IR: 1736 cm^{-1}

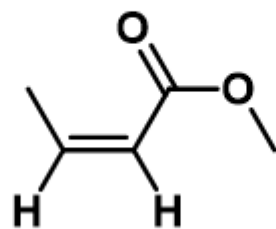
1H NMR (δ , ppm): 3.59 (s, 3H), 3.32 (t, 2H), 2.25 (t, 2H), 1.85–1.75 (m, 2H), 1.73–1.62 (m, 2H)

^{13}C NMR (δ , ppm): 174.0, 51.0, 32.9, 32.8, 31.0, 23.0

(CSIR NET)

100. Reaction of styrene ($\text{PhCH}=\text{CH}_2$) with HBr gives a mixture of regioisomers **A** (major) and **B** (minor). The ^1H NMR spectrum of the mixture shows four signals, amongst others, at δ 5.17, 3.53, 3.15 and 2.00 ppm with relative integration of 2:1:1:6, respectively. The molar ratio of **A** and **B** is

Q3. Predict the number of signals expected in the ^1H NMR spectrum of the following compound and assign order of chemical shift (δ) values to each type of proton.
(CSIR NET)



¹³C NMR spectra

❑ **Broad-band Decoupling:** Decouples all the coupling interactions in a molecule

➤ Advantages:

1. Spectra become simpler
2. The intensity of the lines increases due to NOE

➤ Disadvantage: We lose information whether a carbon is C, CH, CH₂ or CH₃

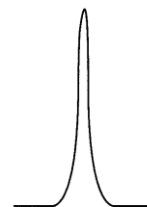
❑ DEPT

➤ **Distortionless Enhancement by Polarization Transfer**

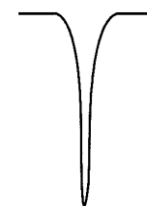
➤ The interactions between spins are manipulated through variable pulse sequence

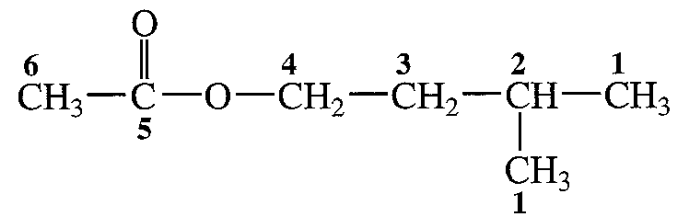
- **DEPT-45**: Only carbons with attached hydrogens are observed; quaternary carbons are not observed, CDCl_3 peak is not observed
- **DEPT-90**: Only carbons which bear single hydrogen are observed
- **DEPT-145**: All carbons having attached hydrogen(s) are observed

CH and CH_3 : positive peak

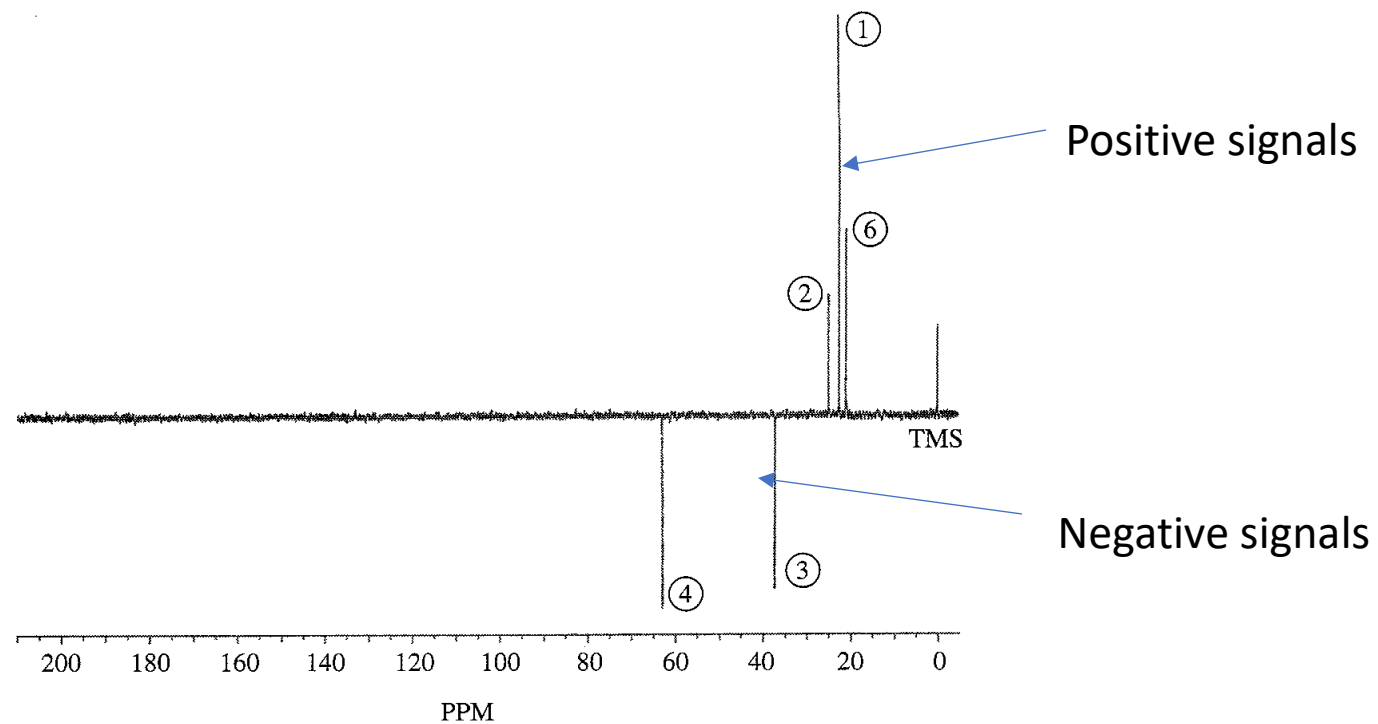


CH_2 : negative peak



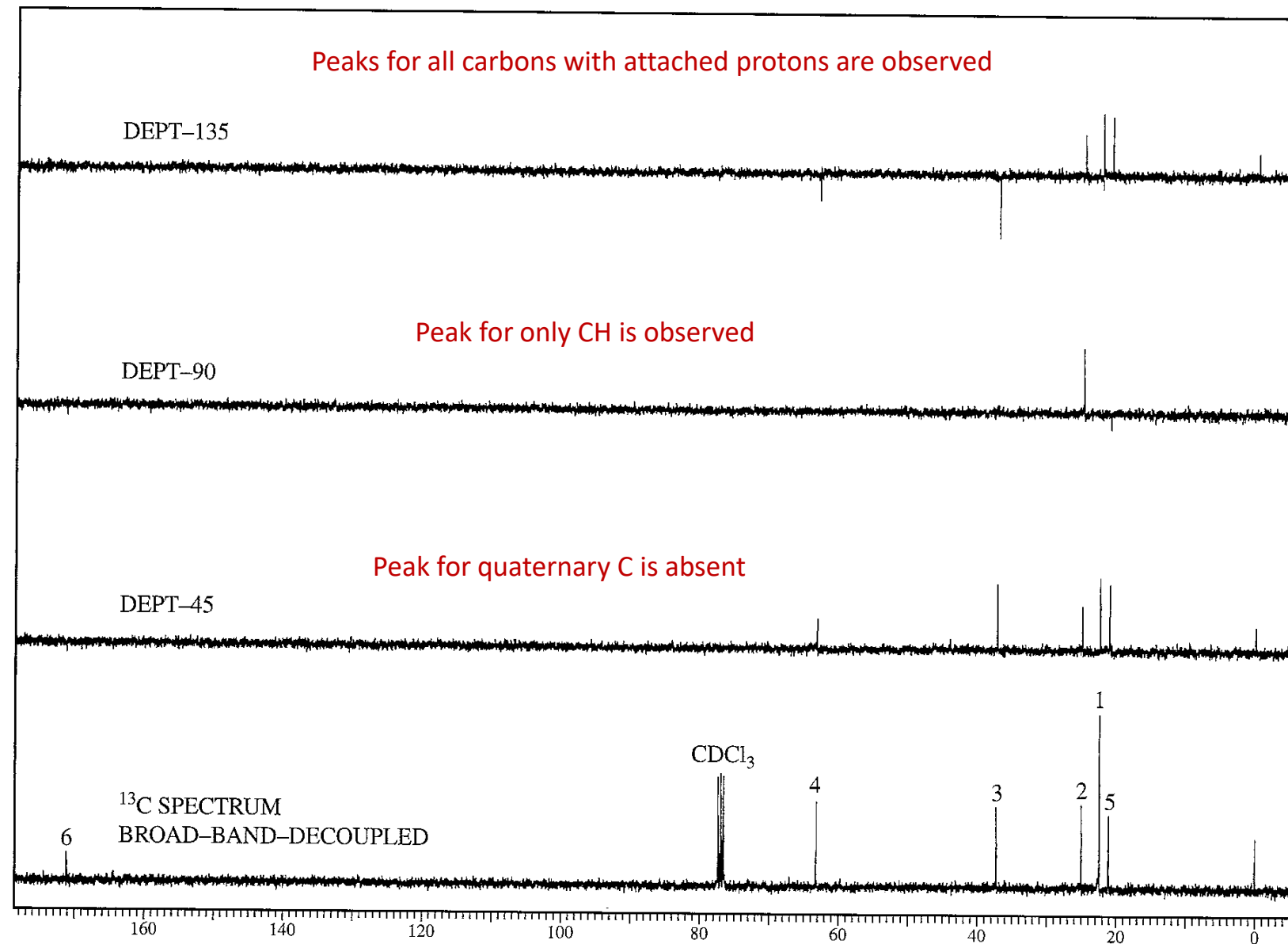
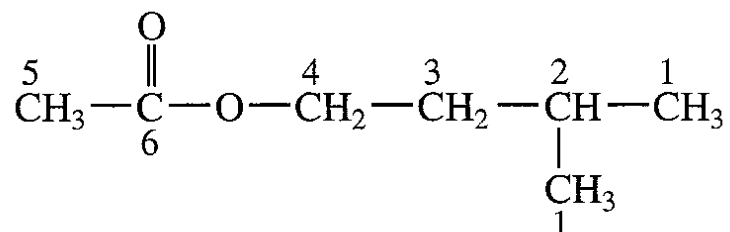


ISOPENTYL ACETATE IN CDCl_3 —DEPT-135

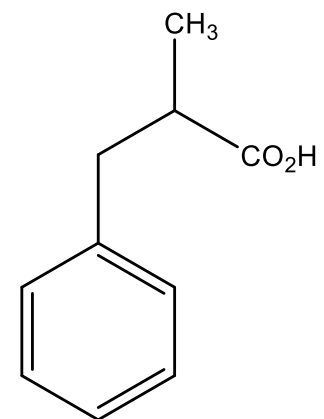


➤ Some of the signals are positive, some of them are negative!

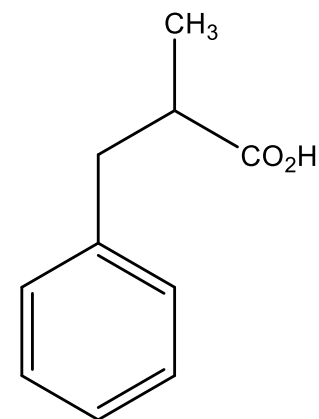
DEPT technique:



Normal Carbon	DEPT-135	DEPT-90
22 ppm		
36		
43		
126.4		
126.6		
128		
145		
179		



Normal Carbon	DEPT-135	DEPT-90
22 ppm	Positive	No peak
36	Positive	Positive
43	Negative	No peak
126.4	Positive	Positive
126.6	Positive	Positive
128	Positive	Positive
145	No peak	No peak
179	No peak	No peak



Q. Isomeric compounds **A** and **B** have the composition $C_{11}H_{12}O_4$. Deduce structures for **A** and **B** from the spectral data (in the proton NMR, ortho and meta couplings are visible). **Compound A**: IR (KBr): 1720 (s), 1240 (s) cm^{-1} , 1H NMR ($CDCl_3$): δ 2.46 (s, 3H), 3.94 (s, 6H), 8.05 (d, J 2 Hz, 2H), 8.49 (t, J 2 Hz, 1H); **Compound B**: IR (Nujol[®]): 1720 (s), 1245 (s) cm^{-1} , 1H NMR ($CDCl_3$): δ 2.63 (s, 3H), 3.91 (s, 6H), 7.28 (d, J 8 Hz, 1H), 8.00 (dd, J 8 Hz, 2 Hz; 1H), 8.52 (d, J 2 Hz, 1H). Assign all protons to the appropriate peaks.

$$DBE: C - (H/2) - (X/2) + (N/2) + 1$$